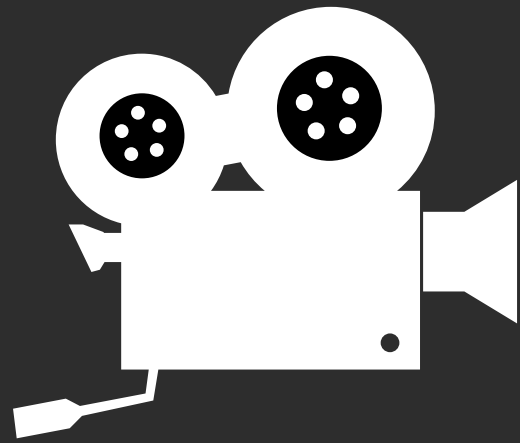


GRP NO - 2

# MOVIE GENRE CLASSIFIER

NLP PBL PHASE 1





# TEAM MEMBERS



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# INTRODUCTION



In film and television, genres are important because they create expectations. Think about it – when someone asks you if you want to see a movie, one of the first questions you will ask is: “What kind of movie?”

Our task is to build a model that can predict the genre of a movie using just the plot details (available in text form). We will use MultiLabel Classification Model.

It is used when there are two or more classes and the data we want to classify may belong to none of the classes or all of them at the same time, e.g. to classify which traffic signs are contained on an image.

# DATASET

We will use [CMU Movie Summary Corpus](#) open dataset for our project. This contains multiple files, but we'll focus on only two of them for now:

- `movie.metadata.tsv`: Metadata for 81,741 movies, extracted from the November 4, 2012 dump of Freebase. The movie genre tags are available in this file
- `plot_summaries.txt`: Plot summaries of 42,306 movies extracted from the November 2, 2012 dump of English-language Wikipedia. Each line contains the Wikipedia movie ID (which indexes into `movie.metadata.tsv`) followed by the plot summary



# INPUTS & OUTPUTS



We will be using Multi-label Classification for movie genre classification.

Inputs will be both the dataset files, steps are:

- Explore the dataset
- Data cleaning
- Data Visualization
- Data preprocessing
- Model building
- Prediction

Output to be expected are:

- Top 50 genre
- most frequent used stopwords visualisation
- Prediction based on features of the summary



# NEED & NOVELTY



**NEED: UNDERSTANDING THE CHALLENGES OF MOVIE  
GENRE CLASSIFICATION**

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**NEED: DATA COLLECTION AND PREPROCESSING**

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**NEED: EXPLORING DIFFERENT FEATURE  
REPRESENTATIONS**

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**NEED: EVALUATION METRICS**

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**NOVELTY: MULTI-LABEL CLASSIFICATION**

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**NOVELTY: FINE-GRAINED GENRE CLASSIFICATION**

# APPLICABILITY



Movies are one of the most prominent means of entertainment. The widespread use of the Internet in recent times has led to large volumes of data related to movies being generated and shared online.

It will help people in choosing which movie to see and which not, not wasting their time much.

# ROLES & TIMELINE



**Yash Akotkar**  
Coding



**Rajdeep Chaurasia**  
Dataset gathering &  
cleaning



**Netal Daga**  
Coding & visualization



**Samarth Ghule**  
Dataset gathering &  
cleaning

Deadline: By 31st March, 2023



# REFERENCES

- [Multi-label Movie Genre Classification Using Multiple Modalities](#)
- [Movie genre classification: A multi-label approach based on convolutions through time](#)
- [Multi-label Movie Genre Detection from a Movie Poster Using Knowledge Transfer Learning](#)

Deadline: By 31st March, 2023



A large audience is seated in a lecture hall, facing a large projection screen. The screen displays the text "THANK YOU FOR LISTENING!" in bold, red, sans-serif capital letters. The audience is seen from behind, filling the foreground and middle ground. The lecture hall has a modern design with blue and grey walls and a dark ceiling. The text is centered on the screen and is the most prominent element in the image.

**THANK YOU FOR  
LISTENING!**